Data wisdom

INTRODUCTION TO DATA

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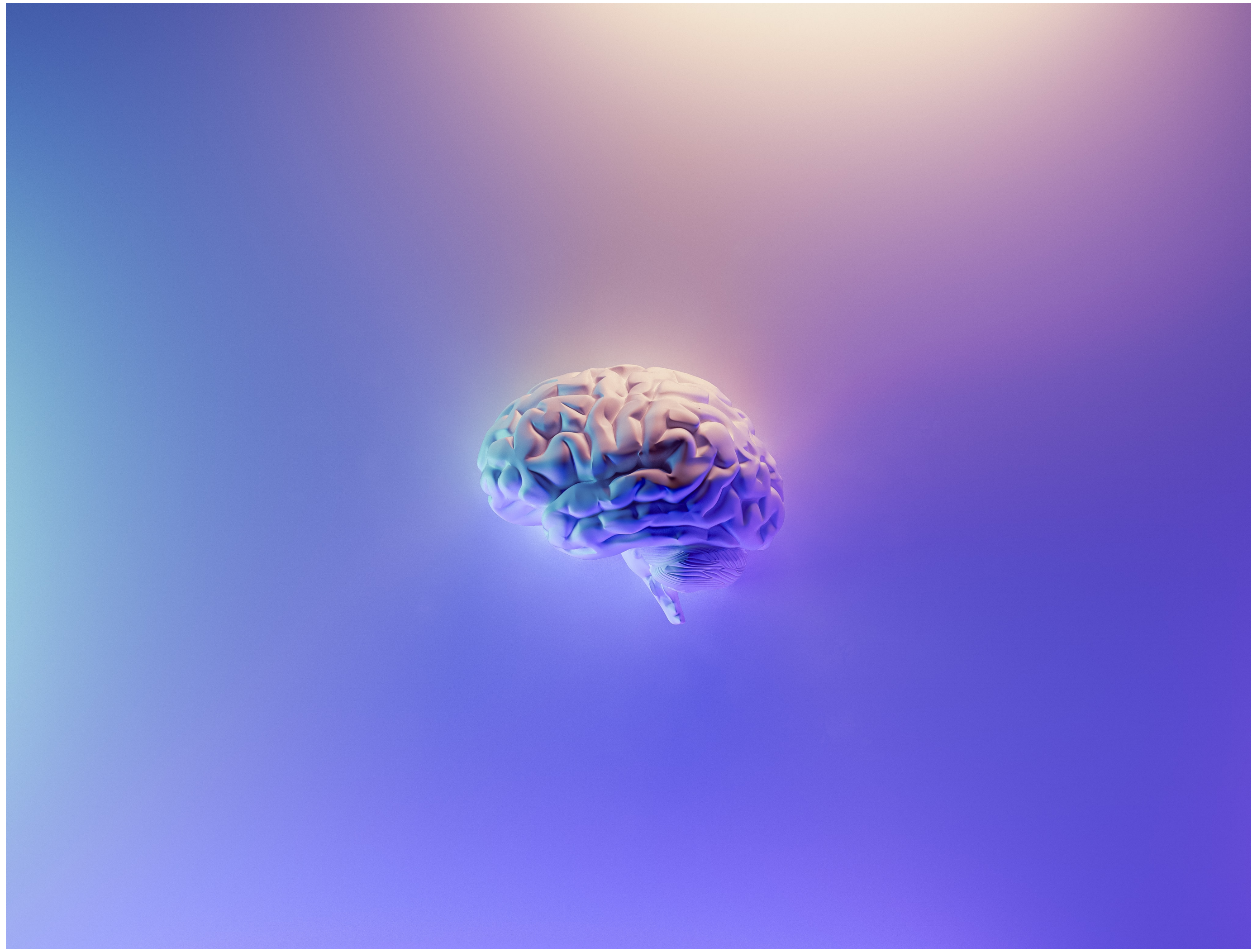
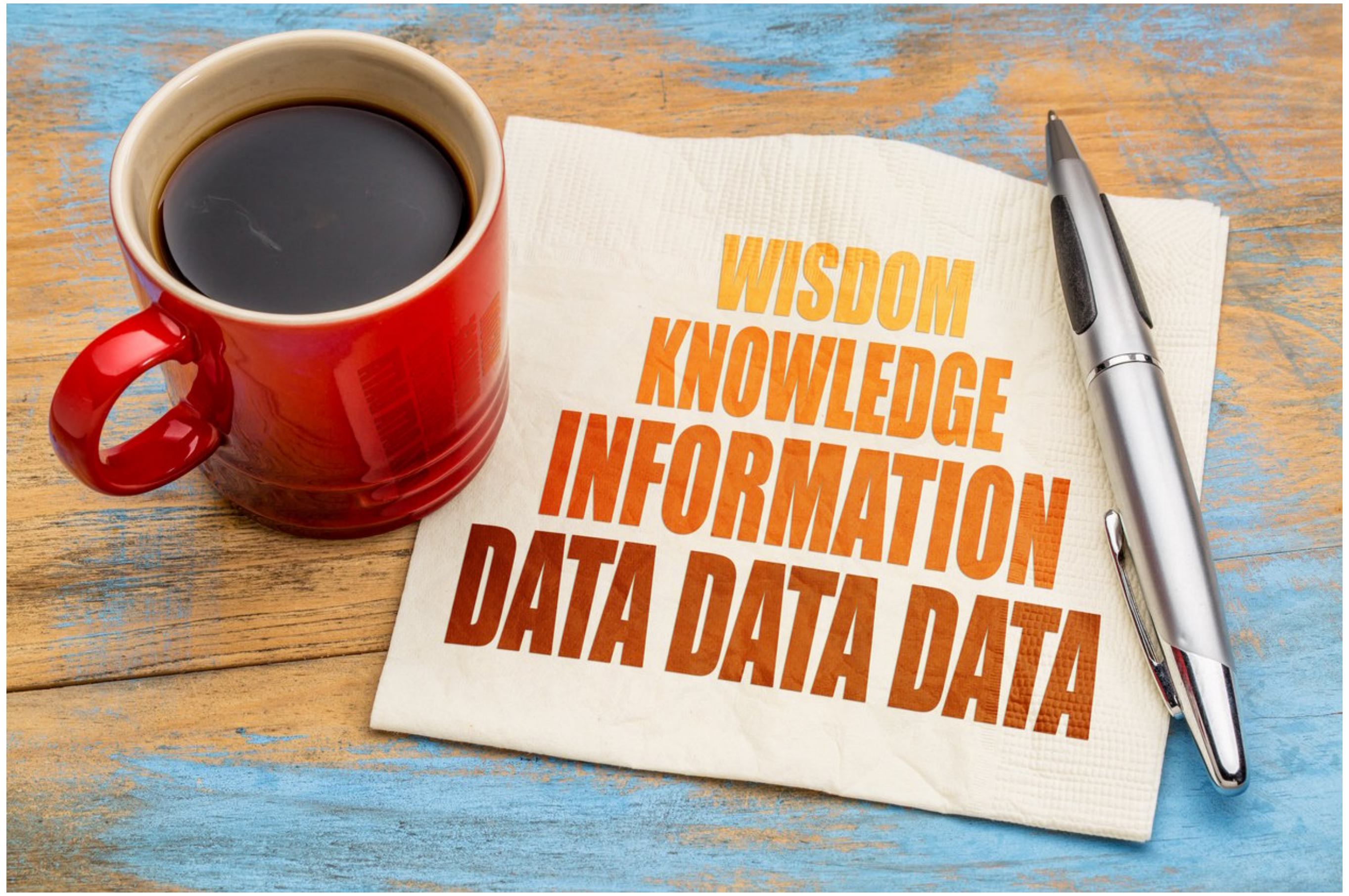
Mental models



Mental models present a framework to understand the vast data landscape surrounding us

The DIKW pyramid



From raw data to valuable decisions

A four-layer model:

1. D - Data
2. I - Information
3. K - Knowledge
4. W - Wisdom

Raw data

Data is the **foundation** of the DIKW Pyramid

- **Raw data**: data without context
- **Example A:** 4, 80%, cloudy, 75
- **Example B:** party, park, 2, 15

Creating information

Information is (organized) data with **context**

Example A:

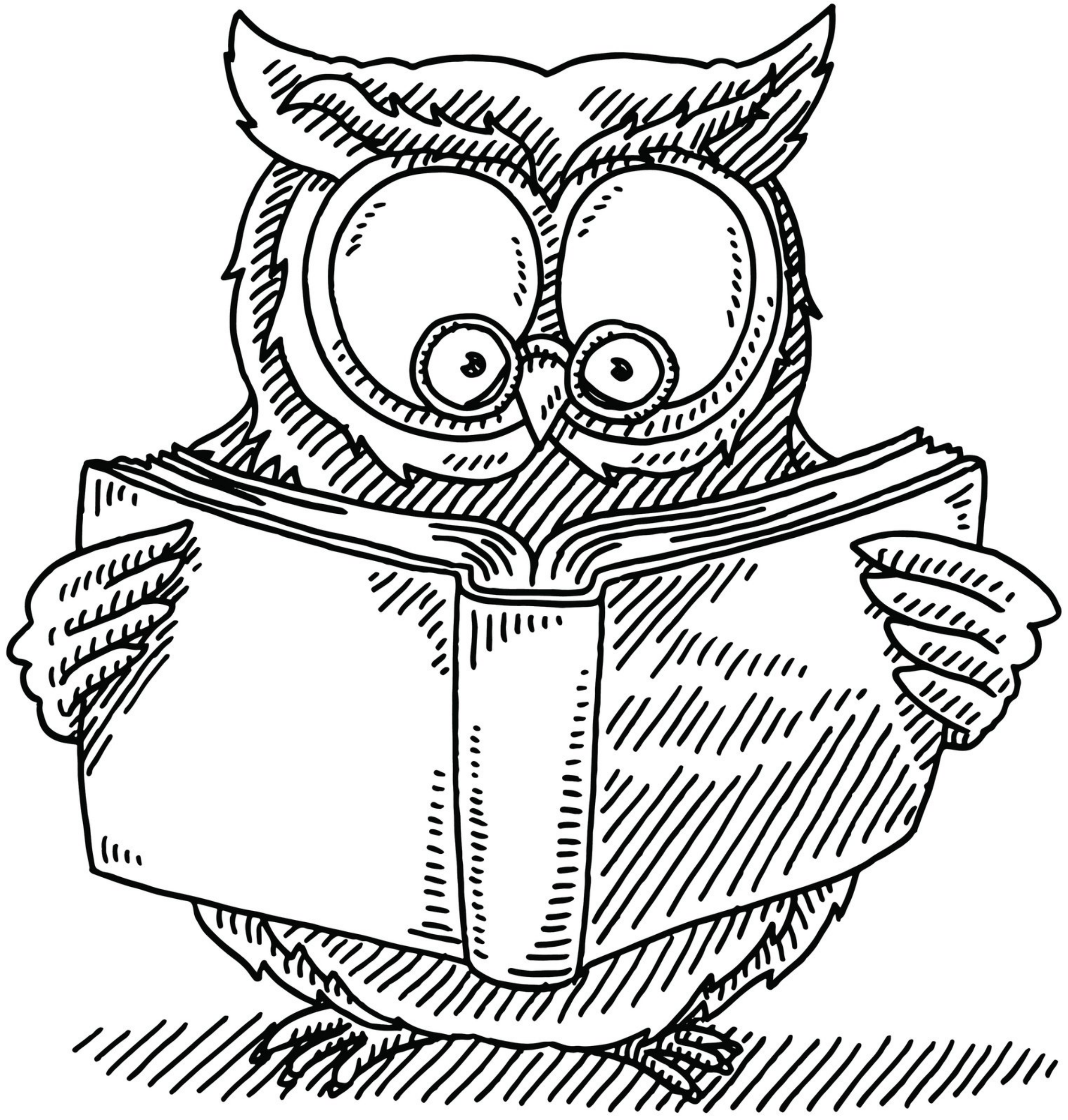
Data:	4	80	cloudy	75
Information:	4 inches	80% chance	cloudy outside	75 degrees

Example B:

Data:	party	park	2	15
Information:	birthday party	park across street	age 2	15 guests

Knowledge is power

- Information alone doesn't lead to decisions
- Connecting all the dots of information
- Knowledge is information with **meaning**



Data:

- Example A: 4, 80%, cloudy, 75
- Example B: party, park, 2, 15

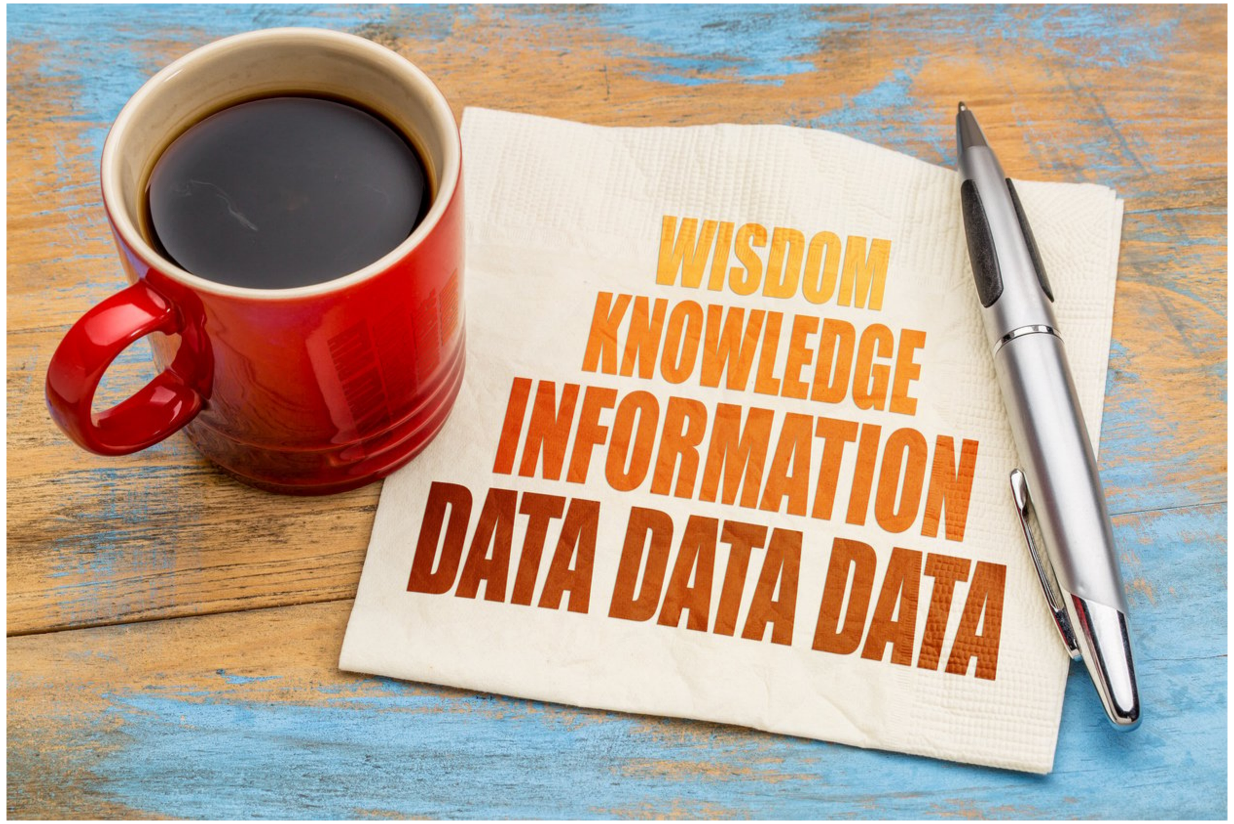
Information:

- 4 inches, 80% chance, cloudy outside, 75 degrees
- Birthday party, park across street, age 2, 15 guests

Knowledge:

- It is currently cloudy and 75 degrees outside, there is an 80% chance of 4 inches of rain tonight. James' second birthday party is tomorrow across the street at the park, there will be 15 guests.

Achieving wisdom



- The **hardest** part
- **Insights:** add more meaning to information by linking pieces
- **Apply** the knowledge for better decisions

Wisdom achieved!

Knowledge:

- It is currently cloudy and 75 degrees outside, there is an 80% chance of 4 inches of rain tonight. James' second birthday party is tomorrow across the street at the park, there will be 15 guests.

Wisdom:

- Because it will probably rain tonight and make the park muddy, James' parents should plan some alternative games for their fifteen guests.

Let's practice!

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Data in decision making

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What is decision-making

- **Minor** decisions:
  - Low impact
  - Related to routine life
- **Major** decisions:
  - High impact
  - Education, residence, financial matters

Decision making: process to make the right **choices** at the right **time**

From data to decision

Asking the right question

A good question will

- Outline exact what you are looking to answer
- Prevent scope creep
- Ensure success throughout the rest of the process

Collecting data

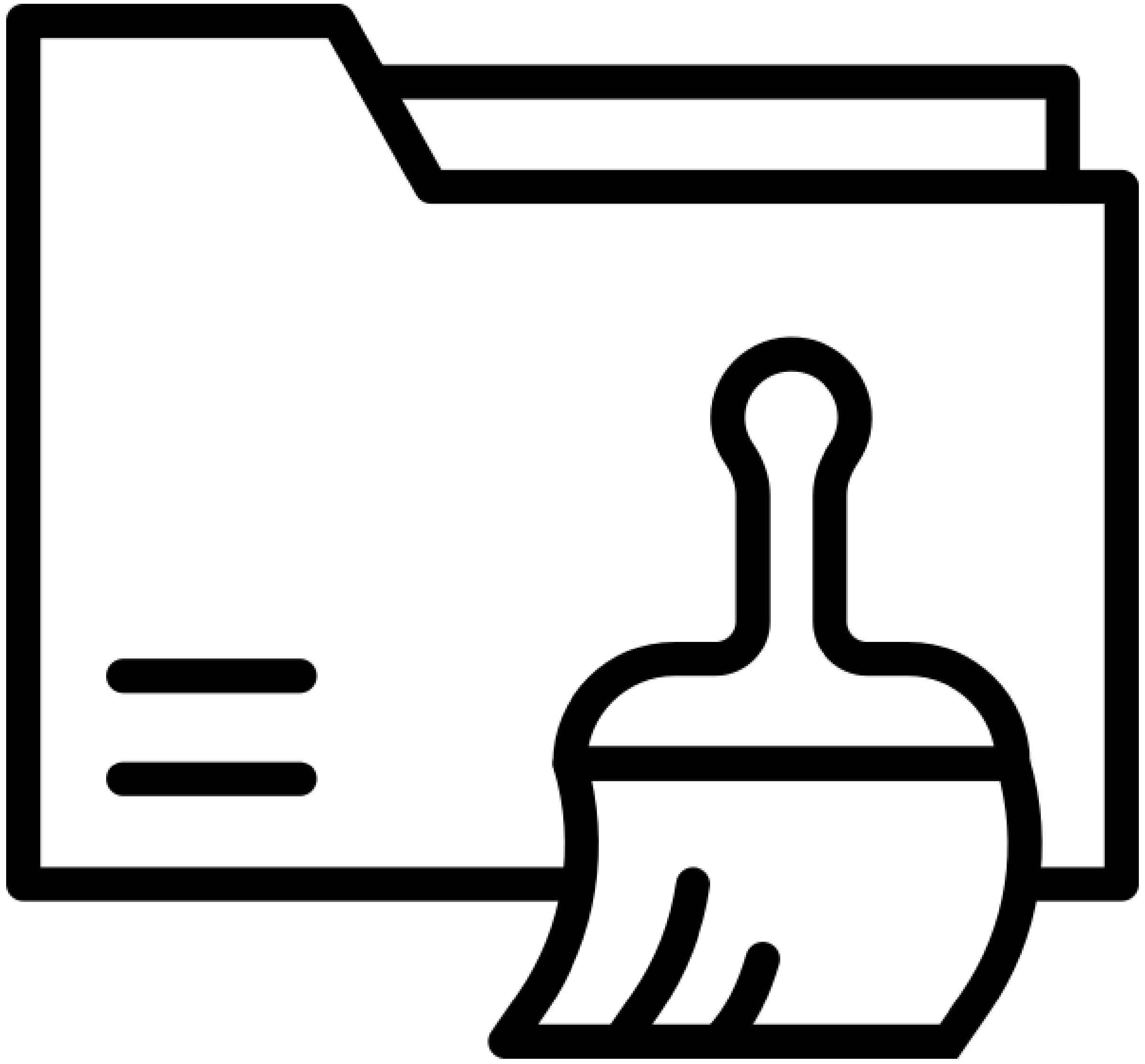
Find the **right** data for your question

- Data can live in multiple locations
- Think ahead to your analysis, you can save time and effort now

Preparing data

Prepare data for analysis:

- Clean bad to good data
- Arrange data into expected structure for analysis
- Sometimes most time consuming task
  - 80% of time spent on data cleaning



Analyzing data

Decisions are based on data analysis

Data analysis tools:

- Python, R
- Tableau, Power BI
- Excel, Google Sheets
- *and many more*

Making decisions

- The last step of the data-driven process
- Better decisions based on data
  - … especially when combined with personal experiences
- Iterative process

Let's practice!

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Data as a resource

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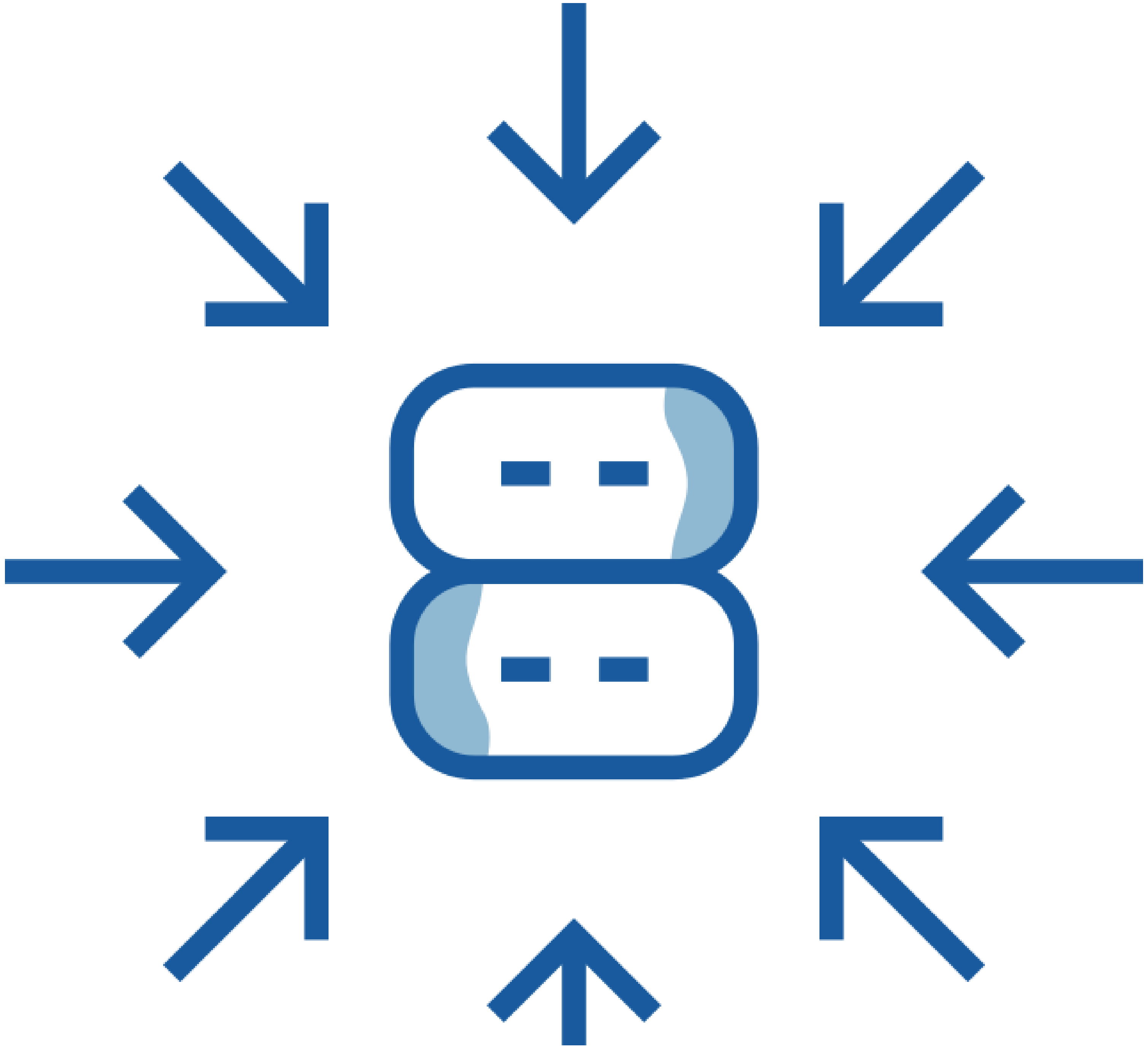
Overwhelming data

Data is often too large in its raw form

- Even "simple" analytics require large amounts of data
- More complex analysis can leverage millions or billions of records

Being able to summarize a dataset into smaller pieces is required to make informed decisions

Summing things up



Aggregations translate raw data into summaries that are easier to understand

Common aggregations:

- Simple average (mean)
- Totals also known as sums
- Minimums and maximums
- Modes

Aggregations allow you to focus on a specific attribute of a dataset

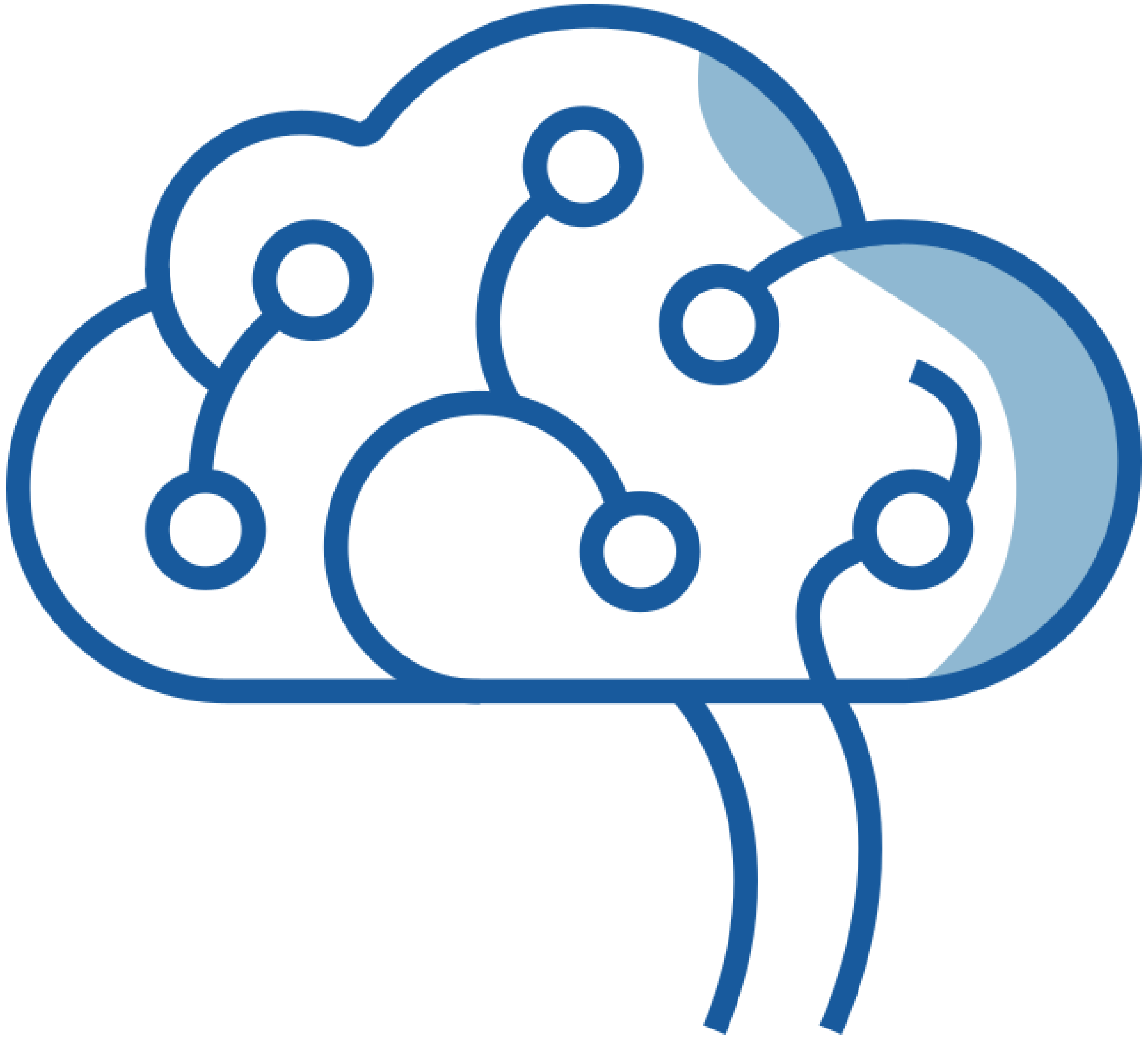
See the big picture

Aggregations appear in many ways throughout organizations

- Metrics
- Benchmarks
- Key Performance Indicators (KPIs)

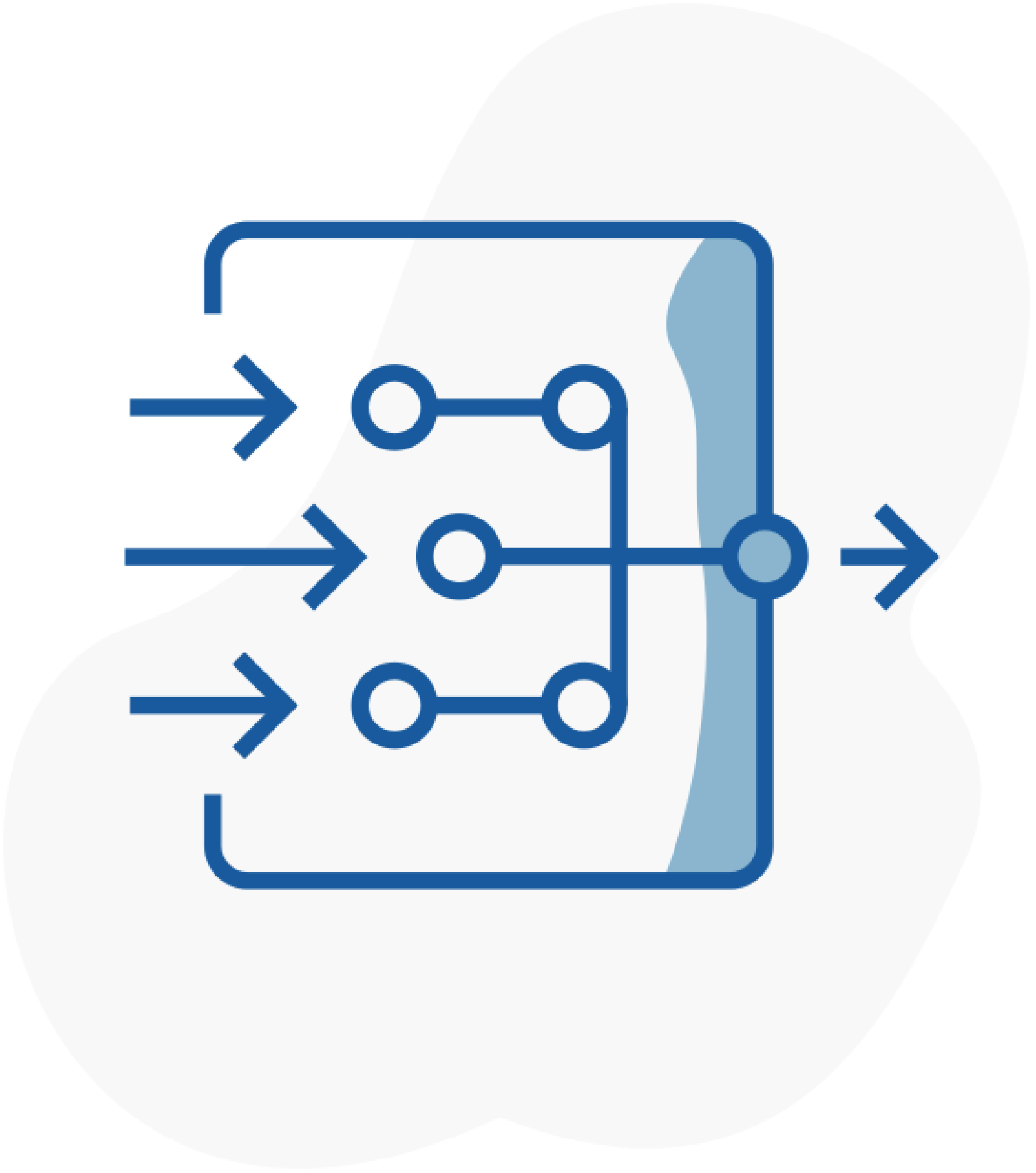
Understanding the how these aggregations are created is extremely helpful for many investigations

Curiosity



Take a moment to ask why?

Data flow



Data flow within organizations is often highly complex

- Data from many different source systems
- Processed through other systems
- Displayed and manipulated in other systems

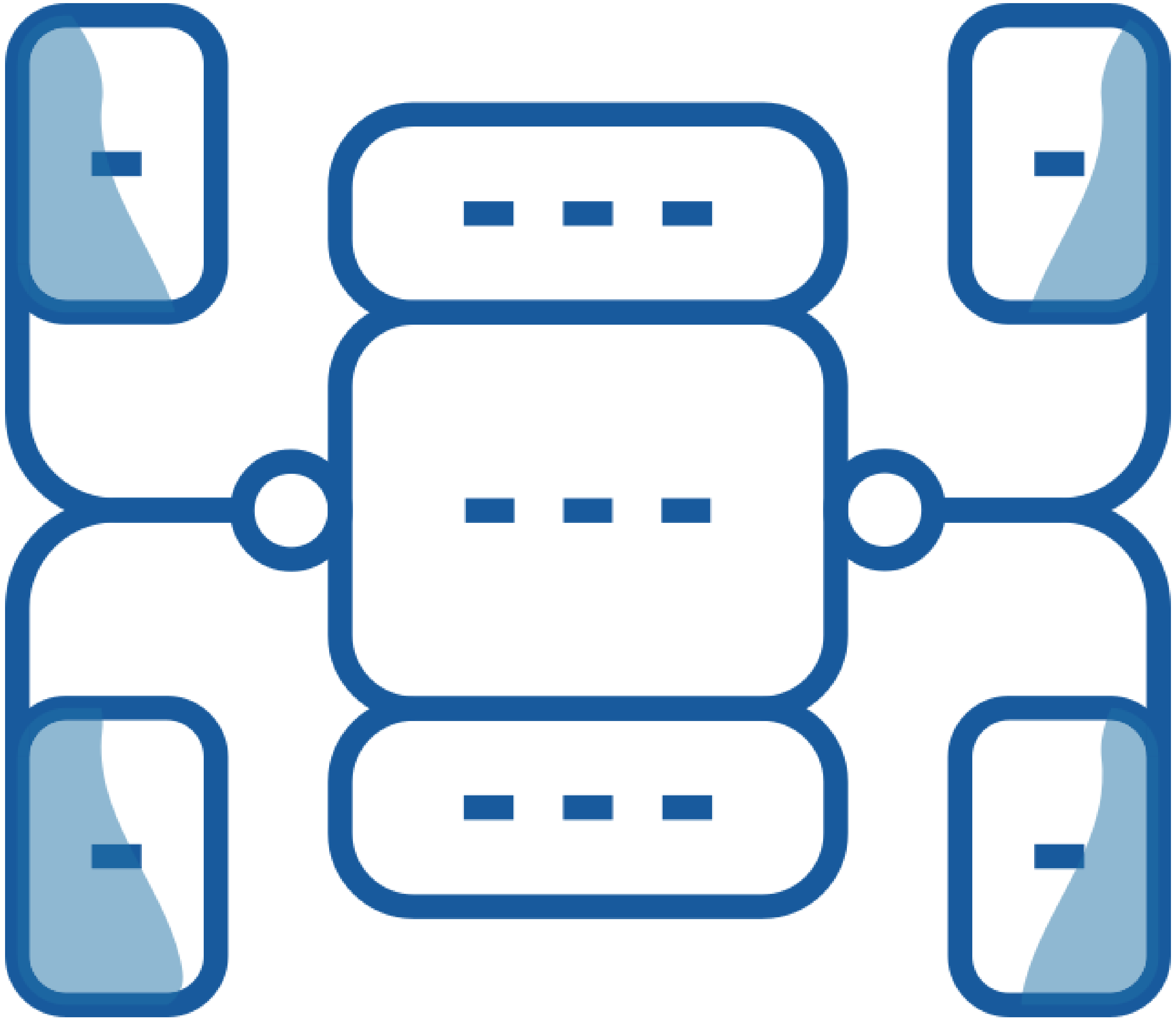
The field of **data management** is responsible for trying to unify all these flows

Data domains

Data Governance: ensure data is consistent, trustworthy and isn't misused

Data Quality: ensure data is accurate, valid, complete and consistent

Data Privacy and Security: ensure proper data access, use and protection



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